Math 210A Notes

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1 Basics

Proposition (Bijection). The map f is a bijection if and only if there exists $g : B \to A$ such that $f \circ g$ is the identity map on B and $g \circ f$ is the identity map on A.

Definition (Binary Relations, Equivalence Relations, Equivalence Classes, Partitions). (1) A **binary relation** on a set A is a subset R of $A \times A$ and we write $a \sim b$ if $(a, b) \in R$.

(2) The relation $\sim$ on A is said to be:

(a) **Reflexive** if $a \sim a$ for all $a \in A$;

(b) **Symmetric** if $a \sim b$, then $b \sim a$ for all $a, b \in A$;

(c) **Transitive** if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.

Proposition (Isomorphism Properties). If $\varphi : G \to H$ is an isomorphism, then

(a) $|G| = |H|$

(b) G is abelian if and only if H is abelian.

(c) for all $x \in G$, $|x| = |\varphi(x)|$.

2 Group Actions

Definition (Group Actions). A **group action** of a group G on a set A is a map $G \times G \to A$ (written $g \cdot a$ for all $g \in G$ and $a \in A$) satisfying the following properties:

(1) $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$ for all $g_1, g_2 \in G$ and $a \in A$;

(2) $1 \cdot a = a$ for all $a \in A$.

Example (Group Actions). (1) The additive group $\mathbb{R}$ that acts on $\mathbb{R}^2$ by $r \cdot (x, y) = (x + ry, y)$.

(2) Let G be any group and let $A = G$. The map $g \cdot a : G \times A \to G$ defined by

$$g \cdot a = gag^{-1} \ \forall g, a \in G$$

satisfy the axioms of a group action.

2.1 Permutation Representation of a Group Action

Let the group G act on the set A. For each fixed $g \in G$, the map $\sigma_g : A \to A$ is a

(1) Permutation for each fixed $g \in G$ and

(2) The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism; that is, the map $\varphi : G \to S_A$ is any homomorphism from a group G to a symmetric group S_A defined by

$$g \cdot a = \varphi(g)(a).$$

3 Section 2.1: Subgroups

Definition (Subgroup). Let G be a group. The subset G of G is a **subgroup** of G if $H \neq \emptyset$ and H is closed under products and inverses; that is, for all $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$. If H is a subgroup of G, we shall write $H \leq G$.

Definition (The Subgroup Criterion). A subset H of a group G is a subgroup if and only if

(1) $H \neq \emptyset$, and

(2) For all $x, y \in H$, $xy^{-1} \in H$.

4 Section 2.2: Centralizers, Normalizers, Stabilizers, and Kernels

4.1 Centralizers, Centers, and Normalizers

Definition (Centralizer). Define $C_G(A) = \{g \in G : gag^{-1} = a \ \forall a \in A\}$. This subset of G is called the **centralizer** of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.

Definition (Center). Define $Z(G) = \{g \in G : gx = xg \ \forall x \in G\}$, the set of elements commuting with all elements of G. This subset of G is called the **center** of G.

Definition (Normalizer). Define $gAg^{-1} = \{gag^{-1} : a \in A\}$. Define the **normalizer** of A in G to be the set $N_G(A) = \{g \in G : gAg^{-1} = A\}$.

- If G is abelian if and only if $Z(G) = G$.
- G is abelian if and only if $C_G(A) = N_G(A)$ for any subset A of G.

4.2 Stabilizers and Kernels of Group Actions

In many cases, we can make our task of proving certain subsets of a group by making it about a problem about whether that subset is equal to either its stabilizer or kernel of a group action. The three main sets mentioned above are just special cases of group actions acting on either a fixed element of some set S or keeping all the elements of S fixed.

Definition (Stablizer). If G is a group acting on a set S and s is some fixed element of S, the **stabilizer** of s in G is the set

$$G_s = \{g \in G : g \cdot s = s\}.$$

Definition (Kernel). Let G be a group acting on a set S. Then the **kernel** of the action of G on S is the set
$$\{g \in G : g \cdot s = s \ \forall s \in S\}.$$

Let $S = \mathcal{P}(G)$ (the power set of G) (the collection of all subsets of G, and let G act on S by *conjugation*), that is, for each $g \in G$ and each $B \subseteq G$ let
$$g : B \to gBg^{-1} \text{ where } gBg^{-1} = \{gbg^{-1} : b \in B\}.$$
This action gives us the stabilizer of A in G; that is,
$$\begin{aligned} N_G(A) &= \{g \in G : gAg^{-1} = A\} \\ &= \{g \in G : g \cdot A = A\} \\ &= G_A \end{aligned}$$
which is clearly a subgroup of G.

5 Cyclic Groups and Cyclic Subgroups

Definition (Cyclic). A group H is **cyclic** if H can be generated by a single element; that is, there exists some element $x \in H$ such that $H = \{x^n : n \in \mathbb{Z}\}$ (where as usual the operation is multiplication).

- If the group in question is additive then we say that $H \subseteq G$ is cyclic if
$$H = \{nx : n \in \mathbb{Z}\}$$
- We say that H is generated by x (or x is the generator of H).
- A cyclic group may have more than one generator.

Proposition. If $H = \langle x \rangle$, then $|H| = |x|$ (where if one side of this equality is infinite, so is the other). More specifically,

(1) If $|H| = n < \infty$, then $x^n = 1$ and $1, x, x^2, \ldots, x^{n-1}$ are all distinct elements of H, and

(2) If $|H| = \infty$, then $x^n \neq 1$ for all $n \neq 0$ and $x^a \neq x^b$ for all $a \neq b$ in $\mathbb{Z}$.

Proposition ((3)). Let G be an arbitrary group, $x \in G$ and let $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $d = (m, n)$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x| \mid m$.

Theorem ((4)). Any two cyclic groups of the same order are isomorphic. More specifically,

(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map
$$\begin{aligned} \varphi : \langle x \rangle &\to \langle y \rangle \\ x^k &\mapsto y^k \end{aligned}$$
is well defined and is an isomorphism.

Proposition ((5))**.** Let G be a group, let $x \in G$ and let $a \in \mathbb{Z} - \{0\}$.

(1) If $|x| = \infty$, then $|x^a| = \infty$.

(2) If $|x| = n < \infty$, then $|x^a| = \dfrac{n}{(n,a)}$.

(3) In particular, if $|x| = n < \infty$ and a is a positive integer dividing n, then $|x^a| = \frac{n}{a}$.

Proposition ((6))**.** Let $H = \langle x \rangle$.

(1) Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.

(2) Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a,n) = 1$. In particular, the number of generators of a H is $\varphi(n)$ (where φ is Euler's φ-function).

This is mainly used to give a description of all the subgroups of H that generate H.

Example (Multiplicative Group)**.** Find all the subgroups of $Z_{20} = \langle x \rangle$.

(1) Find all the divisors of 20. That is, the divisors of 20 are

$$\{1, 2, 4, 5, 10, 20\}.$$

(2) From the above proposition (2) the divisors of 20 gives us the unique subgroups of $\langle x \rangle$ that generate Z_{20}. These subgroups being

$$\langle x \rangle, \langle x^2 \rangle, \langle x^4 \rangle, \langle x^5 \rangle, \langle x^{10} \rangle, \langle x^{20} \rangle.$$

(3) Using prop (5), we can find the order of each subgroup:

$$\begin{aligned}
|\langle x^2 \rangle| &= \frac{20}{(2,20)} = \frac{20}{2} = 10 \text{ elements in } \langle x^2 \rangle \\
|\langle x^4 \rangle| &= \frac{20}{(4,20)} = \frac{20}{4} = 5 \text{ elements in } \langle x^4 \rangle \\
|\langle x^5 \rangle| &= \frac{20}{(5,20)} = \frac{20}{5} = 4 \text{ elements in } \langle x^5 \rangle \\
|\langle x^{10} \rangle| &= \frac{20}{(10,20)} = \frac{20}{10} = 2 \text{ elements in } \langle x^{10} \rangle \\
|\langle x^{20} \rangle| &= \frac{20}{(20,20)} = \frac{20}{20} = 1 \text{ element in } \langle x^{20} \rangle
\end{aligned}$$

(4) List out the generators of each subgroup by computing the Euler-Totient function $\varphi(n)$. For example, given the cyclic subgroup $\langle x^2 \rangle$ which has order 10, we have

$$\varphi(10) = \varphi(5 \cdot 2) = 4 \text{ generators}.$$

Note that φ gives us the number of $a \in \mathbb{Z}^+$ up to 10 such that $(a, 10) = 1$ (where a and 10 are relatively prime). Indeed, we have

- $(1, 10) = 1 \Longrightarrow (x^2)^1 = x^2$ is a generator;
- $(2, 10) = 2 \Longrightarrow (x^2)^2 = x^4$ is NOT a generator;
- $(3, 10) = 1 \Longrightarrow (x^2)^3 = x^6$ is a generator;
- $(4, 10) = 2 \Longrightarrow (x^2)^4 = x^8$ is NOT a generator;

- $(5,10) = 5 \implies (x^2)^5 = x^{10}$ is NOT a generator;
- $(6,10) = 2 \implies (x^2)^6 = x^{12}$ is NOT a generator;
- $(7,10) = 1 \implies (x^2)^7 = x^{14}$ is a generator;
- $(8,10) = 2 \implies (x^2)^8 = x^{16}$ is NOT a generator;
- $(9,10) = 1 \implies (x^2)^9 = x^{18}$ is a generator;
- $(10,10) = 10 \implies (x^2)^{10} = x^{20}$ is NOT a generator;

(5) To describe the containments of each of these subgroups in relation to each other, we fix each divisor in our list and identify which ones can be divided by it. For instance, we see that

$$\begin{aligned} 1 &\mid 2,4,5,10,20 \\ 2 &\mid 4,10,20 \\ 4 &\mid 20 \\ 5 &\mid 10,20 \\ 10 &\mid 20. \end{aligned}$$

This gives us the following containments:

$$\begin{aligned} &\langle x^{20}\rangle \subseteq \langle x^{10}\rangle \subseteq \langle x^5\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^{10}\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^{4}\rangle \subseteq \langle x^2\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^{2}\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x\rangle. \end{aligned}$$

Theorem. Let $H = \langle x\rangle$ be a cyclic group.

(1) Every subgroup of H is cyclic. More precisely, if $K \le H$, then either $K = \{1\}$ or $K = \langle x^d\rangle$, where d is the smallest positive integer such that $x^d \in K$.

(2) If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a\rangle \neq \langle x^b\rangle$. Furthermore, for every integer m, $\langle x^m\rangle = \langle x^{|m|}\rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of H correspond with the integers $1,2,3,\ldots$.

(3) If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d\rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m\rangle = \langle x^{(n,m)}\rangle$ so that the subgroups of H correspond bijectively with the positive divisors of n.

6 Week 6

6.1 Cyclic

Proposition (8). If $\mathcal{A}$ is any collection of subgroups of a group G, then $\bigcap_{H\in\mathcal{A}} H$ is a subgroup of G.

Proof. This can be shown pretty easily. ■

Definition (The Generating Set of a Group). If A is any subset of the group G define

$$\langle A\rangle = \bigcap_{A\subseteq H, H\leq G} H$$

where A is called the **generating set**.

Notice that in the notation of the above proposition, we have

$$\mathcal{A} = \{H \leq G : A \subseteq H\}.$$

Notice that $\mathcal{A} \neq \emptyset$ as G is an element of $\mathcal{A}$ since $G \leq G$ and $A \subseteq G$.

Proposition. $\langle A\rangle$ is the unique minimal element of $\mathcal{A}$.

Proof. Since $A \subseteq H$ for all $H \in \mathcal{A}$, $A \subseteq \langle A\rangle$. Then $\langle A\rangle \in \mathcal{A}$. Let $K \in \mathcal{A}$. Note that

$$\bigcap_{H\in\mathcal{A}} H \leq K.$$

That is, $\langle A\rangle \leq K$. Hence, $\langle A\rangle$ is the minimal element with respect to inclusion. ■

When A is finite, that is,

$$A = \{\alpha_1, \alpha_2, \ldots, \alpha_n\} \text{ for some } n \in \mathbb{N},$$

we write

$$\langle A\rangle = \langle a_1, a_2, \ldots, a_n\rangle.$$

This is a more concrete version of the previous set $\langle A\rangle = \bigcap_{H\leq G, K\subseteq H} H$. Define the following set

$$\overline{A} = \{a_1^{\varepsilon_1} a_2^{\varepsilon_2} a_3^{\varepsilon_3} \ldots a_n^{\varepsilon_n} : n \in \mathbb{N}, \varepsilon_i = \pm 1\ \forall i\}.$$

If $A = \emptyset$, define $\overline{A} = \{1\}$.

Proposition. $\langle A\rangle = \overline{A}$

6.2 Quotient Groups and Homomorphisms

Let G be a group and $N \leq G$. Define a relation on G by

$$a \sim b \iff a^{-1}b \in N.$$

It is straightforward to prove that this defines an equivalence on relation on G. For $a \in G$, the equivalence class of a is

$$\begin{aligned}\{b \in G : a \sim b\} &= \{b \in G : a^{-1}b \in N\} \\ &= \{b \in G : a^{-1}b = n \text{ for some } n \in N\} \\ &= \{b \in G : b = an \text{ for some } n \in N\} \\ &= \{an : n \in N\} \\ &:= aN.\end{aligned}$$

Definition (Cosets). Suppose $N \leq G$ where G is a group and $g \in G$. Let

$$gN = \{gn : n \in N\}$$
$$Ng = \{ng : n \in N\}$$

be called a **left coset** and **right coset** of N in G, respectively. Any element of a coset is called a

representative of a coset.

We will denote the set of all left cosets of N (or right cosets) in G by G/N (read as G mod N).

Proposition (4). Let $N \leq G$ be a group and G/N forms a partition of G. For all $a, b \in G$, $aN = bN$ if and only if $a^{-1}b \in N$. In particular, $aN = bN$ if and only if a and b are representatives of the same coset.

Proof. Since we have recognized left cosets as equivalence classes induced by an equivalence relation, they form a partition, that is,

$$G = \bigcup_{g \in G} gN$$

and $g_1N = g_2N$ if and only if $g_1N \cap g_2N \neq \emptyset$ for all $g_1, g_2 \in G$.

($\Longrightarrow$) Suppose $a^{-1}b \in N$, then

$$a^{-1}b = n \text{ for some } n \in N.$$

It follows that $b = an \in aN$. Since N is a subgroup, $1 \in N$. Hence, $b \cdot 1 \in bN$. It follows that

$$aN \cap bN \neq \emptyset \Longrightarrow aN = bN.$$

($\Longleftarrow$) Suppose that $aN = bN$. Then $an = b$ for some $n \in N$. Hence, $n = a^{-1}b \in N$. We have

$$\begin{aligned} an = bn &\Longleftrightarrow a^{-1}b \in N \\ &\Longleftrightarrow b \in aN \text{ and } a \in aN \\ &\Longleftrightarrow a \text{ and } b \text{ are representatives of } aN \text{ (or } bN). \end{aligned}$$

■

Proposition (5).
(1) The operation on G/N described by $aN \cdot bN = (ab)N$ for all $a, b \in G$ is well defined if and only if $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.
(2) If the operation above is well-defined, then G/N forms a group, where $1N$ is the identity and $(gN)^{-1} = g^{-1}N$.

Proof.
(1) ($\Longleftarrow$) (We will show that the operation defined above is well-defined) To show that the operation is well-defined, we need to show that

$$(ab)N = (a_1b_1)N.$$

Suppose that $gng^{-1} \in N$ for all $g \in G$ and $n \in N$. Let $a, a_1 \in aN$ and $b, b_1 \in bN$. Then $a_1 = an$ and $b_1 = bm$ for some $n, m \in N$. To get our result, we need to show that $a_1b_1 \in (ab)N$ if and only if $(a_1b_1)N = (ab)N$.

Now, we will prove the former. Indeed, we have

$$\begin{aligned} a_1b_1 &= (an)(bm) \\ &= a(bb^{-1})nbm \\ &= ab(b^{-1}nb)m. && (b^{-1}n(b^{-1})^{-1} \in N) \end{aligned}$$

Hence, it follows that $a_1b_1 = (ab)n_1m$ where $n_1 \in N$ and $n_1 = b^{-1}n(b^{-1})^{-1}$.

($\Longrightarrow$) This direction is left for the reader to prove.

(2) This is left for the reader to prove. ■

7 Week 7

Last time we constructed cosets from an equivalence relation and looked at the set of all left cosets of subgroup N of a group G, denoted by

$$G/N = \{gN : g \in G\}.$$

We showed that the operation on G/H defined by

$$(aH)(bH) = (ab)N$$

is well-defined when $gng^{-1} \in H$ for all $g \in G$ and for all $h \in N$. This makes G/N a group. We will prove that this operation is associative, the rest is left to the reader.

Let $N \leq G$ a group where "coset multiplication" is well-defined. Let $aN, bN, cN \in G/N$ $(a, b, c \in G)$. Then

$$\begin{aligned} aN(bNcN) &= aN((bc)N) \\ &= (a(bc))N \\ &= ((ab)c)N \\ &= ((ab)N)cN \\ &= (aNbN)cN. \end{aligned}$$

Similarly, we can show that $1N$ is the identity of G/N and show that

$$(aN)^{-1} = a^{-1}N \quad \forall aN \in G/N.$$

Indeed, for every $gN \in G/N$

$$gN = (1_G g)N = (1_G N) \cdot (gN)$$

and

$$gN = (g1_G)N = (1_G N) \cdot (gN).$$

Now, we have that since $g \in G$ and $g^{-1} \in G$, it follows that

$$1_G N = (gg^{-1})N = (gN) \cdot (g^{-1}N)$$

and

$$1_G N = (g^{-1}g)N = (g^{-1}N) \cdot (gN).$$

This tells us that G/N is a group where N has the nice property of being "normal". This will be expanded in the following definition:

Definition (Normal Subgroups of G). A subgroup N of G is called **normal** in G if every element of g normalizes N. That is, N is normal in G if

$$gNg^{-1} = N \quad \forall g \in G.$$

If N is a normal subgroup of G, then we write $N \trianglelefteq G$.

Theorem (6). Let N be a subgroup of G. The following are equivalent

(1) $N \trianglelefteq G$

(2) $N_G(N) = G$

(3) $gN = Ng$ for all $g \in G$

(4) The operation coset multiplication is well-defined.

(5) $gNg^{-1} \subset N$ for all $g \in G$.

Why does (5) only need containment? It is because you can construct a bijective map between gNg^{-1} and N.

Example (Checking if $\langle s \rangle$ is normal in D_16). We need to examine gsg^{-1} for an arbitrary $g \in D_{16}$. Note that

$$D_{16} = \{1, r, \ldots, r^7, s, sr, \ldots, sr^7\}.$$

Letting $g = s^i r^j$ where $i \in \{0,1\}$ and $j \in \{0,1,2,\ldots,7\}$. Then we have

$$\begin{aligned} gsg^{-1} &= (s^i r^j)s(s^i r^j)^{-1} && (s^i = s^{-i}) \\ &= s^i r^j s r^{-j} s^i \end{aligned}$$

Note that when $i = 0$, we have

$$\begin{aligned} gsg^{-1} &= r^j s r^{-j} \\ &= r^j r^j s \\ &= r^{2j} s. \end{aligned}$$

When $j = 1$, this gives us $gsg^{-1} = r^2 s$. But note that $r^2 s \notin \langle s \rangle$ because otherwise r^2 is either the identity or s which r^2 is neither of these two which is a contradiction. Hence, gsg^{-1} cannot be in $\langle s \rangle$ and so it cannot be a normal subgroup of D_16.

Theorem (Big!). A subgroup N of a group G is normal if and only if it is the kernel of some homomorphism.

Proof. ($\Longrightarrow$) Suppose $N \trianglelefteq G$. Let's define the mapping $\pi : G \to G/N$ by $\pi(g) = gN$ for all $g \in G$. Notice that since N is a normal subgroup of G, this map is well-defined. Let $g_1, g_2 \in G$. Then

$$\pi(g_1 g_2) = (g_1 g_2)N = (g_1 N)(g_2 N) = \pi(g_1)\pi(g_2).$$

Hence, π is a homomorphism. It remains to show that $\ker(\pi) = N$. Indeed, we have that

$$\begin{aligned} \ker(\pi) &= \{g \in G : \pi(g) = 1N\} \\ &= \{g \in G : gN = 1N\} \\ &= \{g \in G : g \in 1N\} \\ &= \{g \in G : g \in N\} \\ &= N. \end{aligned}$$

Hence, $\ker(\pi) = N$.

($\Longleftarrow$) This direction will be on homework. ■

Definition (Natural Projection). Let N be normal in G. The homomorphism π from the previous theorem is called the **natural projection** (homomorphism) of G onto G/N. If $\overline{G} \leq G/N$, the complete pre-image of $\overline{H}$ is $\pi^{-1}(\overline{H})$.

Remark. If $\overline{H} \subseteq G/N$, then $N \leq \pi^{-1}(\overline{H})$ since

$$1N \in \overline{H} \Longrightarrow N = \ker(\pi) = \pi^{-1}(1N) \subseteq \pi^{-1}(\overline{H}).$$

Example (Complete Pre-image of Q_8). $\pi^{-1}(\overline{H}) = \{1, -1, i, -i\}$ where

$$\pi(i) = i\langle -1\rangle$$
$$\pi(-1) = -1\langle -1\rangle = 1\langle -1\rangle$$

7.1 More on Normal Subgroups

There are a lot of ways to see if a subgroup is normal.

Remark (Important). Let G be a group. Observe that $\{1\} \trianglelefteq G$ and $G \trianglelefteq G$ and $G/\{1\} \cong G$ and $G/G \cong \{1\}$.

- When G is clearly an additive group, we denote left and right cosets $g + N$ and $N + g$, respectively where $N \leq G$.
- When G is abelian, every subgroup is normal; that is, $N \leq G$ where

$$\forall n \in \mathbb{N} \quad gng^{-1} = gg^{-1}n = n \in \mathbb{N} \;\; \forall g \in G.$$

7.2 More on Cosets/Lagrange's Theorem

We move away from normal subgroups and just analyze subgroups.

Theorem (Lagrange's Theorem). If G is a finite group, and H is a subset of G, then

$$|H| \mid |G|$$

and the number of left cosets of H in G is $\frac{|G|}{|H|}$.

The idea of the proof of this theorem is as follows: (see problem 18 and 19 from section 1.7 in the book)

Proof. Recall that left cosets form a partition of G; that is,

$$G = \bigcup_{g \in G} gH.$$

There is a bijection from H to gH and is defined by $h \mapsto gh$ and so

$$|H| = |gH|.$$

At this point, we can conclude that $|G| = k|H|$ where k is the number of distinct left-cosets of H in G. This gives us

$$k = \frac{|G|}{|H|}.$$

■

Definition (Index of a Subgroup). If G is a group (possibly infinite) and H is a subgroup of G, then the number of distinct left cosets of H in G is called the **index of H in G** which is denoted by $|G : H|$ (or $[G : H]$).

Corollary (Corollary to Lagrange's Theorem). If G is a finite group and $x \in G$, then $|x| \mid |G|$.

Proof. We proved that $|x| = |\langle x\rangle|$ where $\langle x\rangle \leq G$. The rest follows from Lagrange's Theorem. ■

Remark. For a finite group G with $H \leq G$, $|G : H| = \frac{|G|}{|H|}$.

Example. Let $G = \mathbb{Z}$ and let $H = 3\mathbb{Z}$, then $|\mathbb{Z} : 3\mathbb{Z}|$ is the number of left cosets of $3\mathbb{Z}$ in $\mathbb{Z}$. Recall that

$$3\mathbb{Z} = \{3x : x \in \mathbb{Z}\}$$

where

$$\begin{aligned} 0 + 3\mathbb{Z} &= 3\mathbb{Z} \\ 1 + 3\mathbb{Z} &= \{1 + 3x : x \in \mathbb{Z}\} \\ 2 + 3\mathbb{Z} &= \{2 + 3x : x \in \mathbb{Z}\} \\ 3 + 3\mathbb{Z} &= 0 + 3\mathbb{Z}. \end{aligned}$$

That is,

$$|\mathbb{Z} : 3\mathbb{Z}| = 3 = |\mathbb{Z}/3\mathbb{Z}|.$$

Corollary. If G is a group of prime order p, then G is cyclic.

Proof. Let $x \in G$ where $x \neq 1$. Then $|x| \mid |G|$ since $|G| = p$ a prime. Hence, $|x| \in \{1, p\}$. Since $x \neq 1$, we have $|x| \neq 1$ and so we would need to have $|x| = p$ and so $\langle x\rangle = G$. ■

Example. A subgroup H of a group G with index $|G : H| = 2$, is normal.

Proof. Let $g \in G - H$, then $gH \neq 1H$. Since $|G : H| = 2$, there are two distinct left cosets of H in G and since one of them is $1H$, the other must be gH. Similarly, there are only two distinct right cosets of H in G; namely, $H1$ and Hg. Since $1H = H1$ and cosets form a partition of G, we have that $gH = G - H = Hg$. Hence, the left and right cosets of H are equal and so H is normal in G. ■

Example (Normal Subgroups of is NOT transitive)**.** Note that the relation "is a normal subgroup of" is not transitive. Let $G = D_8$. Note that the $|D_8| = 8$ and $|\langle s\rangle| = 2$, $|\langle s, r^2\rangle| = 4$. Clearly, we have

$$\langle s\rangle \subseteq \langle s, r^2\rangle \subseteq D_8.$$

We have

$$|D_8 : \langle s, r^2\rangle| = 2$$

and $|\langle s, r^2\rangle : \langle s\rangle| = 2$. Thus,

$$\langle s\rangle \trianglelefteq \langle s, r^2\rangle \trianglelefteq D_8$$

but $\langle s\rangle$ is NOT normal in D_8. Note that $rs \neq sr$ but $rs = sr^{-1}$ where r^2s is not 1 and not s.

Definition. Let $H, K \leq G$ where G is a group. Define

$$HK = \{hk : h \in H,\ k \in K\}.$$

Theorem (Big Theorem of this lecture)**.** If H and K are finite subgroups of a group, then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

where HK need not be a group for the above to hold.

Proof. Notice that HK is the union of left cosets of K; that is,

$$HK = \bigcup_{h \in H} hK.$$

Since each coset of K has $|K|$ elements, we will count the number of distinct cosets in the above union. We know $h_1 k = h_2 k$ for $h_1, h_2 \in H$ if and only if $h_2^{-1} h_1 \in K$. It follows that

$$\begin{aligned} h_1 k = h_2 k &\iff h_2^{-1} h_1 \in K \cap H \\ &\iff h_1(K \cap H) = h_2(K \cap H). \end{aligned}$$

Thus, the number of distinct cosets of the form hK with $h \in H$ is the same as the number of distinct cosets of $K \cap H$ in H. Since we have that $K \cap H$ is a subgroup of H, this is $\frac{|H|}{|K \cap H|}$. Therefore, HK consists of $\frac{|H|}{|H \cap K|}$ distinct cosets of K of which contains $|K|$ elements, it follows that

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

■

Example (An example where HK is not a subgroup)**.**

8 Week 8

8.1 Monday

Recall that a lot of the problems given in the homework are based on more general theory of the results that we have been going thus far. For instance, one can prove that If G is cylic and φ is a homomorphism from G to H, then $\varphi(G)$ is cyclic. Note that $\pi : G \to G/N$ where $g \mapsto gN$ and π is a homomorphism. Another problem we can do is prove that if G is cyclic, then G/N is cyclic.

For an $\psi : G \to H$ where ψ is a group homomorphism. If $|x| = n < \infty$, then $|\psi(x)||x|$. Because π is a homomorphism from $G \to G/N$. Hence, the homework problem 2 follows immediately.

Last time we saw that for $H, K \leq G$ we have

$$|HK| = \frac{|K||H|}{|K \cap H|}.$$

Note that HK may not always be a subgroup of G.

Example. Let $S_3 = G$ and $H = \langle(1\ 2)\rangle$ and $K = \langle(2\ 3)\rangle$. Then

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{2 \cdot 2}{1} = 4.$$

However, we see from Lagrange's theorem that if $HK \leq G$, then $4 \mid 3!$ which is a contradiction. We can further deduce that

$$\langle(1\ 2), (1\ 3)\rangle = S_3.$$

Since $4 \leq |\langle(1\ 2), (2\ 3)\rangle| \leq 6$ and must also divide 6. So, $(1\ 2)$ and $(2\ 3)$ generates all of S_3.

Proposition. If H, K are subgroups of a group G, $HK \leq G$ if and only if $HK = KH$. *Note that $HK = KH$ does not indicate that the elements of the set commute with each other. Only that $hk \in HK$, we have that $hk = k_1 h_1$ for some $h_1 \in H$ and $k_1 \in K$*

Proof. Assume that $HK = KH$. Our goal is to show that HK is a subgroup of G. Since H and K are non-empty, then $HK \neq \emptyset$. It remains to show that given $a, b \in HK$ that $ab^{-1} \in HK$. To this end, let $a, b \in HK$. Then $a = h_1 k_1$ and $b = h_2 k_2$. Then

$$\begin{aligned} ab^{-1} &= (h_1 k_1)(h_2 k_2)^{-1} \\ &= (h_1 k_1)(k_2^{-1} h_2^{-1}) \end{aligned}$$

Since $K \leq G$, we have $k_1 k_2^{-1} = k_3 \in K$ and $h_2^{-1} = h_3 \in H$. This gives that

$$ab^{-1} = h_1 k_3 h_3.$$

Since $HK = KH$, we know that $k_3 h_3 \in HK$. That is, $k_3 h_3 = h_4 k_4$ for some $h_4 \in H$ and $k_4 \in K$. So,

$$ab^{-1} = h_1 h_4 k_4$$

and letting $h_1 h_4 = h_5$ which is in H by closure, we have that

$$ab^{-1} = h_5 k_4 \in HK.$$

Conversely, suppose that $HK \leq G$. Our goal is to show that $HK = KH$; that is, $HK \subseteq KH$ and $KH \subseteq HK$. Since $K \leq HK$ and $H \leq HK$ and so $KH \leq HK$ by closure of HK. To show that $HK \leq KH$, let $hk \in HK$. Since HK is a subgroup, then hk is the inverse of some element $a \in HK$. That is, $hk = a^{-1} = (h_1 k_1)^{-1} = k_1^{-1} h_1^{-1} \in KH$. It follows that $HK \leq KH$. Furthermore, $HK = KH$. ■

Example. Let $G = D_8$, $H = \langle r \rangle$ and $K = \langle s \rangle$. Notice that $rs \in HK$ and $rs = sr^{-1} \in KH$. By the way,

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{4 \cdot 2}{1} = 8.$$

So, $HK = D_8 = KH$.

Corollary. If H and K are subgroups of a group G and $H \leq N_G(K)$, then HK is a subgroup of G. In particular, if $K \trianglelefteq G$, then HK is a subgroup of G for all $H \leq G$. *Note: If $K \trianglelefteq G$, then $N_G(K) = G$ and certainly $H \subseteq N_G(K)$ for any $H \leq G$*

Proof. We will prove that $HK = KH$. Let $h \in H$ and $k \in K$. By assumption, we have that

$h \in N_G(K)$; that is, $hkh^{-1} \in K$. Hence, $hk = hk(h^{-1}h)$ where $hkh^{-1} \in K$ and $h \in H$. Thus, $hk \in KH$. Thus, $HK \subseteq KH$. Similarly, we want to show that $kh = hh^{-1}kh$ where $h \in H$ and $h^{-1}kh \in K$ since $h \in N_G(K)$ implies that $h^{-1} \in N_G(K)$. It follows that $KH \subseteq HK$ and hence $HK = KH$. ■

Theorem. Let H, K be subgroups of a group G with $H \leq K \leq G$. Then the index

$$|G : H| = |G : K||K : H|.$$

Proof. Let g_i be a distinct representative for a left coset of H in G for all $i \in I$ where I is some indexing set. So, we have

$$\{g_iH : i \in I\} = G/H = \{gH : g \in G\}.$$

We have $g_iH = g_jH$ if and only if $g_i = g_j$ (we are counting the elements of gH because we don't know if it will be finite or infinte). *At this point we are invoking the axiom of choice.* Let $\psi : I \times K/H \to G/H$ by

$$\psi(i, kH) = g_ikH.$$

To see that this is well-defined, let us suppose that $k_1H = k_2H$ for some k_1 and k_2 in K. That is, $k_1^{-1}k_2 \in H$. Then $\psi(i, k_1H) = g_ik_1H$ and $\psi(i, k_2H) = g_ik_2H$. So,

$$\begin{aligned}(g_ik_1)^{-1}(g_ik_2) &= (k_1^{-1}g_i^{-1})(g_iK_2)\\ &= k_1^{-1}k_2 \in H\end{aligned}$$

by assumption. Hence, ψ is well-defined. We now show that ψ defines a bijection. Suppose that

$$\begin{aligned}&\psi(i, k_1H) = \psi(j, k_2H)\\ &\implies (g_ik_1)H = (g_jk_2)H\\ &\implies (g_ik_1)^{-1}(g_jk_2) \in H.\end{aligned}$$

This implies that $k_1^{-1}g_i^{-1}g_jk_2 = h$ (call this (*)) for some $h \in H$. This tells us that $g_i^{-1}g_j = k_1hk_2^{-1}$ for some $h \in H$. Note that $H \leq K \leq G$ and so $g_i^{-1}g_j \in K$ since $H \subseteq K$. Then $g_iK = g_jK$. Thus, we have $g_i = g_j$. This further implies that $i = j$. Using this in (*) gives

$$\begin{aligned}&k_1^{-1}h_i^{-1}g_jk_2 = h\\ &\implies k_1^{-1}k_2 = h \in H\\ &\implies k_1H = k_2H.\end{aligned}$$

Hence, we see that φ is one-to-one. Let $gH \in G/H$. Since the left cosets of k partition the group G, we have that $g \in g_iK$. That is, $g = g_ik$ for some $k \in K$. Hence, $\varphi(i, kH) = g_ikH = gH$. Thus, ψ is surjective. Hence,

$$\underbrace{|G : H|}_{G/H} = \underbrace{|G : K|}_{I} \cdot \underbrace{|K : H|}_{K/H}.$$

■

Theorem (First Isomorphism Theorem). If $\varphi : G \to H$ is a homomorphism of groups, then

$$\ker(\varphi) \trianglelefteq G,$$

and

$$G/\ker(\varphi) \cong \varphi(G).$$

Proof. *Proof is left to the reader.* Take $\mu : G/\ker(\varphi) \to \varphi(G)$ where $gK \mapsto \varphi(g)$ and prove that is well-defined and is a bijection. ■

End of Monday Lecture

8.2 Wednesday

Recall that the main goal for this subject is to be able to classify groups. One way we have been gaining information about a certain group is by multiplying two subgroups together. In general, we are trying to develop tools to be able to better classify certain groups.

Theorem (First Isomorphism Theorem)**.** If $\varphi : G \to H$ is a homomorphism of groups, then

$$\ker(\varphi) \trianglelefteq G,$$

and

$$G/\ker(\varphi) \cong \varphi(G).$$

Proof. *Proof is left to the reader.* Take $\mu : G/\ker(\varphi) \to \varphi(G)$ where $gK \mapsto \varphi(g)$ and prove that is well-defined and is a bijection. ■

Corollary (17 from book)**.** Let $\varphi : G \to H$ be a group homomorphism

(1) φ is injective iff $\ker(\varphi) = 1$

(2) $|G : \ker(\varphi)|$

Theorem (Second Isomorphism Theorem)**.** Let G be a group, let A and B be subgroups of G. Also assume that A is contained in the normalizer $N_G(B)$. Then $AB \leq G$ and B is normal in AB, $A \cap B$ is normal in A, and

$$AB/B \cong A/(A \cap B).$$

(The second isomorphism theorem is often called the Diamond Isomorphism Theorem)

Proof. Since $A \leq N_G(B)$ implies $AB \leq G$ and $B \leq N_G(B)$, and $B \leq AB$, it follows that $AB \leq N_G(B)$, and so $B \trianglelefteq AB$. Since $B \trianglelefteq AB$, then AB/B is a group. Define $\varphi : A \to AB/B$ by $\varphi(a) = aB$ for all $a \in A$. Let $a_1, a_2 \in A$. Hence,

$$\varphi(a_1a_2) = (a_1a_2)B = a_1Ba_2B = \varphi(a_1)\varphi(a_2)$$

since AB/B is a group. Notice that $abB = aB$ since $b \in B$. That is, for all $abB \in AB/B$

$$abB = aB.$$

Hence, our mapping is surjective. Hence, $\varphi(A) = AB/B$. Notice that

$$\begin{aligned}\ker(\varphi) &= \{a \in A : \varphi(a) = 1B\}\\ &= \{a \in A : aB = 1B\}\\ &= \{a \in A : a \in B\}\\ &= A \cap B.\end{aligned}$$

Now, we have $\varphi(A) = AB/B$ and $\ker(\varphi) = A \cap B$. Thus, by the first isomorphism theorem, we have

$$A/\ker(\varphi) \cong \varphi(A) \Longrightarrow A/(A \cap B) \cong AB/B$$

where $A \cap B \trianglelefteq A$. ■

Theorem (Third Isomorphism Theorem). Let G be a group, and let H and K be normal subgroups of G with $H \leq K$. Then

$$(G/H)/(K/H) \cong G/K.$$

Proof. Since $K \trianglelefteq G$, we have $\pi_H(K) = \{kH : k \in K\}$, we know that

$$\pi_H(K) \trianglelefteq \pi_H(G) = G/H.$$

Notice that

$$\{kH : k \in K\}$$

Now, let's define our mapping $\varphi : G/H \longrightarrow G/K$ by $gH \mapsto gK$. Now, we want to show that φ is well-defined. Suppose that $g_1H = g_2H$ if and only if $g_1 = g_2h$ for some $h \in H$. Our goal is to show that $\varphi(g_1H) = \varphi(g_2H)$; that is, we want to show that $g_1K = g_2K$. Since $H \leq K$, so $h \in K$ as well; that is, $g_1 = g_2h$ where $h \in K$. So, $g_1K = g_2K$. Hence, $\varphi(g_1H) = \varphi(g_2H)$ and our mapping is well-defined. Next, we will show that this mapping is a homomorphism. Let $g_1H, g_2H \in G/H$ and

$$\varphi(g_1Hg_2H) = \varphi(g_1g_2H) = g_1g_2K = (g_1K)(g_2K) = \varphi(g_1H)\varphi(g_2H).$$

Hence, φ is a homomorphism which is clearly surjective. Hence,

$$\varphi(G/H) = G/K.$$

Finally, we will show that $\ker(\varphi) = K/H$. Indeed, we see that

$$\begin{aligned}
\ker(\varphi) &= \{gH \in G/H : \varphi(gH) = 1K\} \\
&= \{gH \in G/H : gK = 1K\} \\
&= \{gH \in G/H : g \in K\} \\
&= K/H.
\end{aligned}$$

Therefore, by the first isomorphism theorem,

$$(G/H)/(K/H) \cong G/K$$

where $\varphi(G/H) = G/K$ and $\ker(\varphi) = K/H$. ■

The next theorem tell us that when there is a normal subgroup of G, there exists a bijection between subgroups of G and subgroups of the quotient group.

Theorem (The Correspondence Theorem). Let G be a group with $N \trianglelefteq G$. Then there is a bijection from $\{A \leq G : N \leq A\}$ to the set of all subgroups of G/N. In particular, every subgroup of $\overline{G} = G/N$ is of the form $A/N = \overline{A}$ for some subgroup A of G containing N. Namely, the complete pre-image of the subgroup of G/N under the natural projection from G to G/N. This bijection has the following properties:

(i) $A \leq B$ if and only if $\overline{A} \leq \overline{B}$; that is, $A/N \leq B/N$.

(ii) If $A \leq B$, then $|B : A| = |B/N : A/N| = |\overline{B} : \overline{A}|$.

(iii) $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$.

(iv) $\overline{A \cap B} = \overline{A} \cap \overline{B}$

(v) $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$ $(A/N \trianglelefteq G/N)$

Every time we have a bar over a group it represents the image of the natural projection. Now, the question is, "how do we construct the bijection form the subgroups of G containing N to the subgroups of G/N?". The answer is the projection map $\pi : G \to G/N$. Let $A \subseteq G/N$ where

$$\pi(A) = \{aN : a \in A\} \leq G/N.$$

That point here is that π will provide our bijection *this is for you to prove.* One can also prove (ii) on your own. The exam will consists of comp problems, using H and K, and one of the properties from the correspondence theorem.